RDFIA 3 - c

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January 2025

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1 Introduction

This lab focuses on estimating the uncertainty of models, a crucial aspect for identifying potential overconfidence in classification tasks. To achieve this, we will first analyze the characteristics of the most uncertain images in a dataset. Then, we will explore failure prediction and out-of-distribution detection, where uncertainty estimation plays a key role.

2 Part 1 - Monte Carlo Dropout on MNIST

Question 1-1

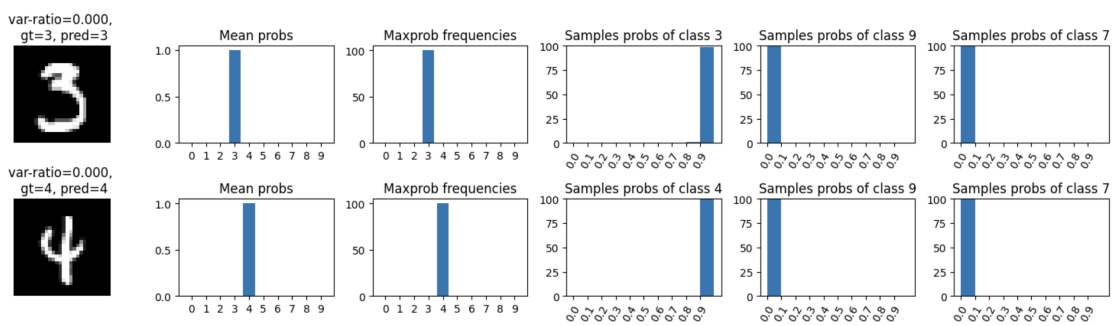

Figure 1: Output probability distribution of 2 random images

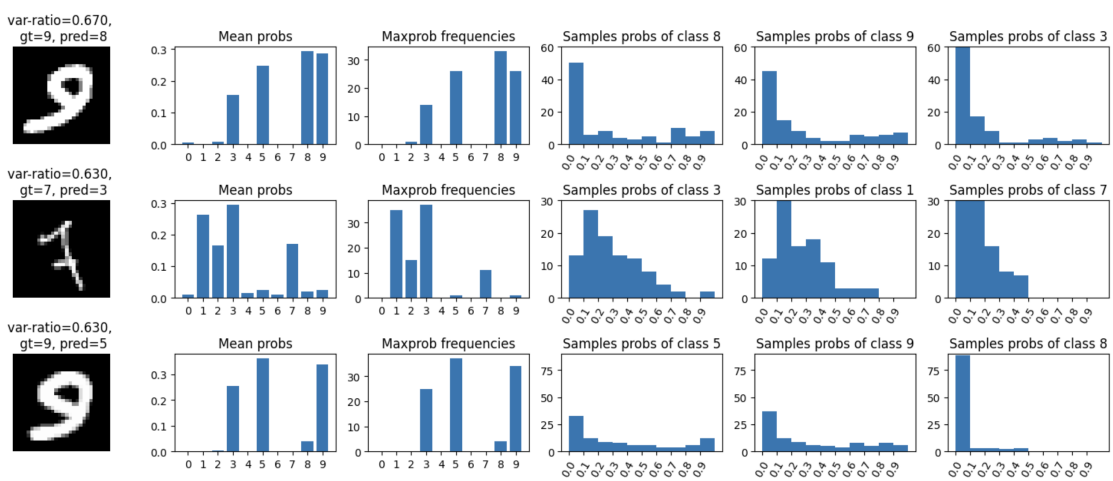

Figure 2: Output probability distribution of the 3 most uncertain images

Firstly, when comparing the images themselves, we can say that those classified with a high level of confidence (FIGURE 1) are very clearly identifiable and show little to no features that could be attributed to another digit. On the other hand, ambiguous images (FIGURE 2 contain patterns that are present in multiple digits, as we can see in the last image: if we ignore the top-right bar that closes the loop of the number 9, it could easily resemble the shape of a 5.

Then, the histograms in FIGURES 1 and 2 show very different characteristics. Those corresponding to well-classified images display a single bar in each histogram, indicating that the model assigns the image to its respective class with a probability very close to 1, while the probability of it being classified elsewhere is nearly 0.

On the other hand, the histograms of ambiguous images are more complex: the first two reflect the model's difficulty in distinguishing between certain classes. For instance, if we take the last image as an example, out of a total of $T = 100$ iterations, the model classifies it as a 5 about 35 times, as a 9 around 30 times, as a 3 roughly 25 times, and as an 8 five times.

Finally, the last three histograms show the distribution of the model's output probabilities for three specific classes highlighting that the ambiguous images are classified with a very low level of confidence since the probabilities are very rarely close to 1. Moreover, it gives information about the confidence of the model: for example the second image of the FIGURE 2 is very often classified with a low probability while the classification's probability of the last one is more uniformly distributed and reaches a high value a few times.

In summary, the uncertainty measures obtained thanks to the variational ratio provide useful information about the model's performance and ability to classify the images. This can highlight for example whether the model is overconfident about certain class or on the contrary if it always classifies certain digit with a low probability.

3 Part 2 - Failure prediction

Question 2-1

To evaluate and compare the methods, we selected an appropriate metric focused on detecting classification errors. These errors are treated as the positive class, while correct predictions are treated as the negative class. Given that our models are highly accurate, classification errors are rare, leading to an imbalanced scenario with many true negatives. The FIGURE 3 shows the precision-recall curves for each method along with their AUPR scores. The results highlight ConfidNet's superior

performance over the other methods, which is likely due to its direct estimation of TCP which is a more reliable uncertainty measure than MCP and entropy. ConfidNet also maintains high precision as recall increases, reflecting its balanced ability to detect true positives while minimizing false positives.

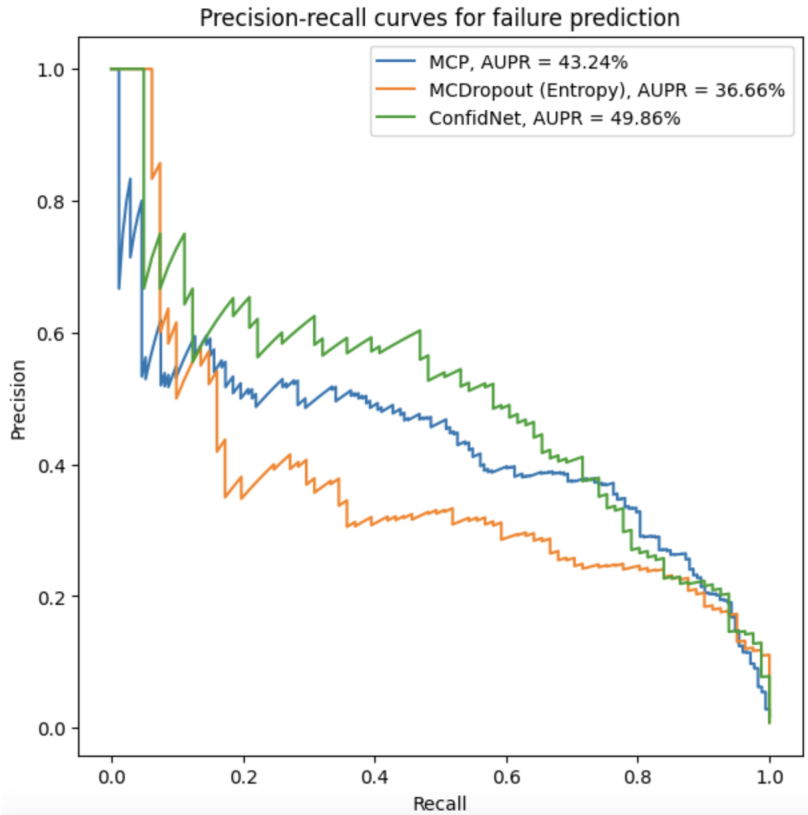

Figure 3: Precision-recall curves for failure prediction for 3 different methods along with their AUPR values

We chose the AUPR (Area Under the Precision-Recall Curve) over the AUROC (Area Under the Receiver Operating Characteristic Curve) because AUROC is less suited for imbalanced datasets. AUROC gives equal weight to both classes, often overstating performance due to the dominance of true negatives. In contrast, AUPR focuses on the minority class, emphasizing precision (the fraction of true positives among predicted positives) and recall (the fraction of true positives identified). This makes AUPR more reliable in our context, where classification errors (the positive class) are far fewer than correct predictions.

4 Part 3 - Out-of-distribution detection

Question 3-1

The FIGURE 4 presents the precision-recall curves for three uncertainty metrics: MCP, ODIN and MC Dropout (MCD) mutual information along with their respective AUPR values. Since the curves are smooth, the analysis primarily focuses on the AUPR scores.

ODIN, which modifies output probabilities, slightly outperforms MCP as expected, given that ODIN builds upon MCP. However, it is unexpectedly outperformed by the MCD-based method: mutual information. Despite also relying on raw output probabilities like ODIN, this method benefits from the stochastic nature of MC Dropout, achieving outstanding AUPR score of 97.32%, making it the most effective metric for out-of-distribution (OOD) detection.

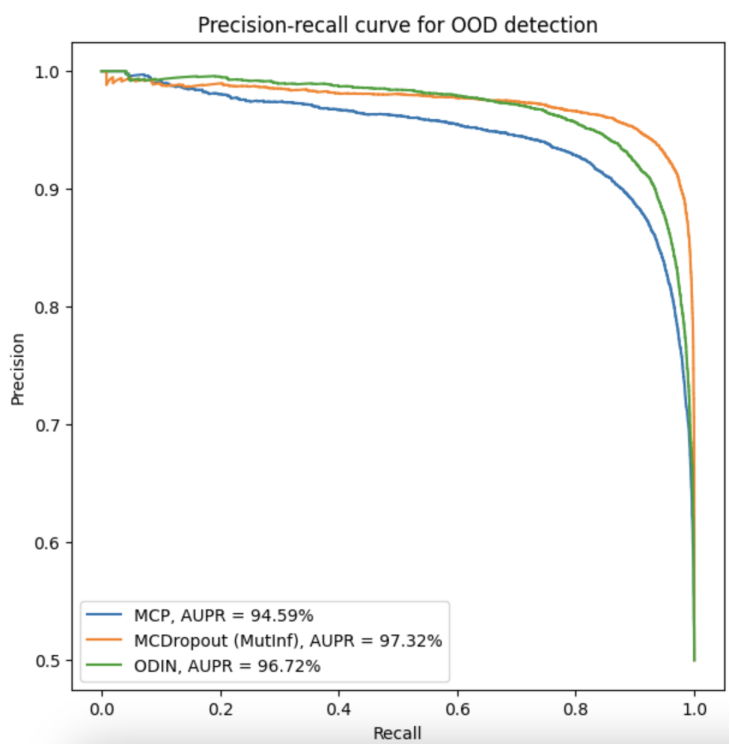

Figure 4: Precision-recall curves for OOD detection for 3 different methods along with their AUPR values